

# Non-small cell lung cancer

Adenocarcinoma · squamous · large cell · **TNM 9th edition (Jan 2025)** · the driver landscape moves faster than any printed card — **re-check NCCN at every new diagnosis**

Compiled from the Stanford Oncology Handbook 2025, NSCLC (Salter A, Roy M) · NCCN · primary trial reports verified against PubMed, September 2026.

## WORKUP

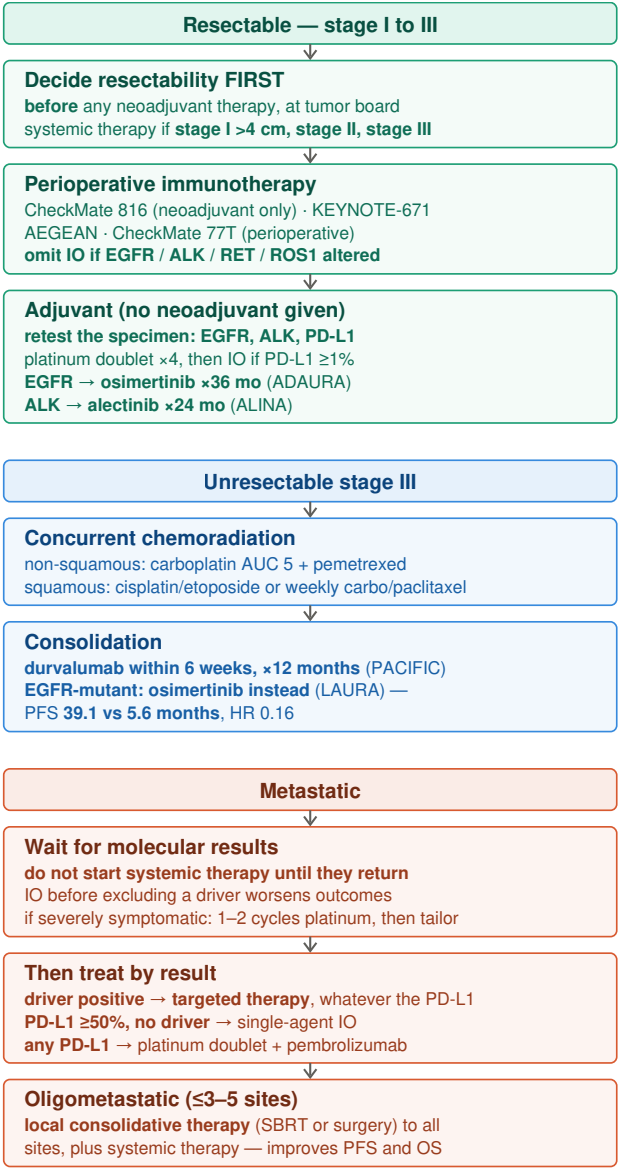
- **Chest CT with contrast** (nodes are far easier to call with contrast)
- **Brain MRI with contrast for every stage except IA** — PET never substitutes for it
- **PET/CT** for distant disease (or CT C/A/P + bone scan if unavailable)
- Biopsy: EBUS, CT-guided transthoracic, metastatic site, or pleural cytology. **Non-decalcified cores preferred for molecular**
- **Pathologic mediastinal nodal evaluation** for non-metastatic disease, especially stage II–III. EBUS reaches 2R/2L, 4R/4L, 7, 10, 11; **mediastinoscopy cannot reach stations 5–6** (needs a Chamberlain procedure)
- **PD-L1 in every case.** Molecular sequencing in all adenocarcinoma; add for squamous if never/light smoker, small biopsy, or adenosquamous
- **Stage IIIB or IV: concurrent tissue + liquid molecular testing** (ctDNA ~7-day turnaround). **A negative liquid biopsy does not rule out a driver** — confirm on tissue

## STAGING — TNM 9TH (JAN 2025)

|                     | Definition                                                                                                                                                  |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>T1 / T2</b>      | ≤3 cm / >3–5 cm, or visceral pleura, adjacent lobe, or main bronchus                                                                                        |
| <b>T3 / T4</b>      | >5–7 cm or chest wall, phrenic nerve, same-lobe nodule / >7 cm or mediastinum, great vessels, carina, esophagus, spine, different ipsilateral lobe          |
| <b>N1 / N2 / N3</b> | Ipsilateral hilar (stations 10–14) / ipsilateral mediastinal or subcarinal — <b>N2a single station, N2b multiple</b> / contralateral or any supraclavicular |
| <b>M1a / M1b</b>    | Pleural-pericardial effusion or nodules, or contralateral lung nodule / <b>single</b> extrathoracic met                                                     |
| <b>M1c1 / M1c2</b>  | Multiple mets, <b>single organ system</b> / <b>multiple organ systems</b>                                                                                   |

What changed from the 8th edition: N2 split into N2a/N2b and M1c split into M1c1/M1c2. T descriptors are unchanged.

## TREATMENT PATHWAY



## SCREENING

- **USPSTF 2021 (Grade B):** annual **low-dose CT**, age **50–80**, ≥20 pack-years, currently smoking or quit within 15 years
- Stop at **15 years since cessation**, or when comorbidity would preclude curative surgery
- **NLST:** 20% reduction in lung-cancer mortality vs chest radiography · **NELSON:** 24% in men (the women’s subgroup did not reach significance)
- **Smoking cessation remains the highest-yield intervention** — duration of smoking matters more than pack count

## HISTOLOGY

| Type                  | Features                                                                                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adenocarcinoma</b> | ~50%; peripheral, non/light smokers. <b>TTF-1, napsin A, CK7</b> positive. <b>Highest yield for actionable drivers.</b> Mucinous (TTF-1–) is KRAS-enriched and EGFR-poor |
| <b>Squamous</b>       | ~25%; central, cavitating, heavy smokers. <b>CK5/6, p40 / p63;</b> keratin pearls and intercellular bridges. <b>Pemetrexed and bevacizumab are both contraindicated</b>  |
| <b>Large cell</b>     | Diagnosis of exclusion — treat as non-squamous                                                                                                                           |
| <b>Adenosquamous</b>  | Both components, each ≥10% of the tumor                                                                                                                                  |

## EARLY STAGE — SURGERY AND RADIATION

- **Lobectomy + mediastinal nodal sampling or dissection** is the standard; VATS preferred over thoracotomy
- **Sublobar resection is acceptable for peripheral cT1aN0 ≤2 cm**, after intraoperative nodal confirmation. **CALGB 140503:** non-inferior DFS (HR 1.01). **JCOG0802:** segmentectomy gave superior 5-yr OS (94.3% vs 91.1%) **but doubled local relapse** (10.5% vs 5.4%), and the survival gain came from non-lung-cancer deaths — **read it as tolerability, not better cancer control**
- **SBRT** for medically inoperable stage I. **Avoid central SBRT within 2 cm of the proximal bronchial tree**
- Preoperative assessment: **predicted postoperative FEV1 and DLCO**; <30% predicted flags high surgical risk

# Non-small cell lung cancer

## DRIVER → THERAPY

| Alteration                                                   | First line and beyond                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EGFR ex19del / L858R</b><br><i>~15% US adeno</i>          | <b>Osimertinib 80 mg daily</b> (FLAURA). <b>Add carbo/pemetrexed ×4</b> for higher-risk disease — CNS involvement, L858R, TP53 co-mutation (FLAURA2: mPFS <b>25.5 vs 16.7 mo</b> ; OS 47.5 vs 37.6). <b>Or amivantamab + lazertinib</b> (MARIPOSA: mPFS <b>23.7 vs 16.6 mo</b> , HR 0.70; <b>OS HR 0.75</b> ) — but grade ≥3 AEs <b>80% vs 52%</b> , and it needs <b>VTE prophylaxis for 4 months</b> |
| <b>EGFR exon 20 ins</b><br><i>~10% of EGFR</i>               | <b>Resistant to osimertinib and the classic TKIs</b> . 1L <b>amivantamab + carboplatin/pemetrexed</b> (PAPILLON: mPFS <b>11.4 vs 6.7 mo</b> , HR 0.40). Post-platinum: <b>sunvozertinib</b> (accelerated, <b>2L+ only</b> ). <b>Mobocertinib was withdrawn</b> in 2023                                                                                                                                |
| <b>ALK</b> <i>~3–7%</i>                                      | <b>Lorlatinib 100 mg daily</b> (CROWN: 5-yr PFS <b>60% vs 8%</b> vs crizotinib, median not reached — the longest single-agent PFS reported in any solid tumor). Watch <b>CNS effects</b> (mood, cognition, seizures) and <b>hyperlipidemia</b> . Alternatives: alectinib, brigatinib, ensartinib                                                                                                      |
| <b>ROS1</b> <i>~1–2%</i>                                     | <b>Four co-preferred 1L agents</b> : crizotinib, entrectinib, repotrectinib, taletrectinib. <b>Prefer entrectinib / repotrectinib / taletrectinib if brain metastases</b> — crizotinib has no CNS activity. Repotrectinib TKI-naïve ORR 79%, mPFS 35.7 mo. <b>Alectinib has no ROS1 activity</b> despite its ALK potency                                                                              |
| <b>RET</b> <i>~1–2%</i>                                      | <b>Selpercatinib</b> — the only randomized 1L evidence (LIBRETTO-431: mPFS <b>24.8 vs 11.2 mo</b> vs chemo ± pembrolizumab, HR 0.47; intracranial ORR 82%). Pralsetinib is an alternative with more pneumonitis. Watch hypertension and transaminitis                                                                                                                                                 |
| <b>MET exon 14</b><br><i>~3–4%, older never-smokers</i>      | <b>Capmatinib</b> (1L ORR 68%) or <b>tepotinib</b> (1L ORR ~57%) — <b>both single-arm, so those two numbers are not comparable</b> . Tepotinib: watch peripheral edema                                                                                                                                                                                                                                |
| <b>BRAF V600E</b><br><i>~2%, smokers</i>                     | <b>Encorafenib + binimetinib</b> (PHAROS, treatment-naïve ORR 75%) or <b>dabrafenib + trametinib</b> (ORR 64%, mPFS 10.9 mo). <b>Pyrexia is the classic combination toxicity</b> — hold and dose-modify                                                                                                                                                                                               |
| <b>KRAS G12C</b><br><i>~13% of adeno</i>                     | <b>Not a first-line target</b> — 1L is chemo-IO. <b>2L after platinum + anti-PD-(L)1: adagrasib</b> (KRISTAL-12: mPFS 5.5 vs 3.8 mo, HR 0.58) or <b>sotorasib</b> (CodeBreak 200: 5.6 vs 4.5 mo, <b>no OS difference</b> ; still on accelerated approval after a failed confirmatory filing — the weakest evidence base on this page)                                                                 |
| <b>HER2 (ERBB2) mutant</b><br><i>~2–3%, non-smoker women</i> | Mostly exon 20 insertions; <b>high rate of brain metastases at presentation</b> . <b>Zongertinib</b> — oral, and as of <b>Feb 2026 it carries a 1L indication</b> (treatment-naïve ORR 76%). <b>T-DXd 5.4 mg/kg q3wk</b> (DESTINY-Lung02: ORR 49%) — <b>ILD in 13–15%, including fatal cases</b> ; monitor with serial imaging                                                                        |
| <b>NTRK fusion</b> <i>&lt;1%</i>                             | Larotrectinib, entrectinib or repotrectinib — all tumor-agnostic approvals                                                                                                                                                                                                                                                                                                                            |
| <b>NRG1 fusion</b>                                           | <b>Zenocutuzumab</b> — accelerated approval, <b>after progression on prior systemic therapy</b> , for NSCLC and pancreatic adenocarcinoma only                                                                                                                                                                                                                                                        |

## PROGRESSION ON OSIMERTINIB

- **Rebiopsy** — resistance is heterogeneous, with no single dominant mechanism
- **On-target**: EGFR **C797S**. **Off-target**: MET amplification, HER2 amplification, acquired fusions
- **Histologic transformation is the one you cannot afford to miss: SCLC transformation** (with RB1 and TP53 loss) or squamous transformation. **Treat SCLC transformation as SCLC** — platinum/etoposide
- **Standard next step is platinum-doublet chemotherapy**
- **Amivantamab + chemotherapy** (MARIPOSA-2: mPFS 6.3 vs 4.2 mo, HR 0.48). **The approved regimen is amivantamab + chemo only** — the lazertinib triplet was not approved, for excess hematologic toxicity
- **Datopotamab deruxtecan** — TROP2 ADC on accelerated approval. **The label is EGFR-mutant-specific**: the all-comers trial met PFS but missed OS

## DRIVER-NEGATIVE METASTATIC

- **PD-L1 ≥50%: pembrolizumab** monotherapy (KEYNOTE-024: OS 30 vs 14 months); atezolizumab or cemiplimab are alternatives. **Use chemo-IO instead if the burden is high or the patient is symptomatic**
- **PD-L1 1–49% or <1%**: chemo-IO doublet. **Non-squamous** → pembrolizumab + carboplatin + pemetrexed (KEYNOTE-189). **Squamous** → pembrolizumab + carboplatin + paclitaxel (KEYNOTE-407)
- **Dual IO** — ipilimumab/nivolumab (CheckMate 227, 9LA) or durvalumab/tremelimumab (POSEIDON): chemo-sparing but more toxic, **most attractive at PD-L1 <1% and in squamous**
- **Second line: docetaxel 35 mg/m² days 1+8** — split dosing is better tolerated than 75 q3wk — ± ramucirumab

## ADJUVANT THERAPY

- **Cisplatin doublet ×4** for stage II–IIIA (LACE: ~5% absolute 5-yr OS gain). **Stage IB only if ≥4 cm or high-risk features; no benefit and possible harm in stage IA**
- **EGFR → osimertinib ×36 months** (ADAURA: DFS HR 0.17, OS HR 0.49). **ALK → alectinib ×24 months** (ALINA: DFS HR 0.24). **Omit IO in both**
- **Atezolizumab** (IMpower010) is approved for **stage II–IIIA with PD-L1 ≥1%**, EGFR/ALK wild-type — but the **2025 update showed no significant OS benefit overall**, with the signal concentrated at PD-L1 ≥50%
- **Pembrolizumab** (KEYNOTE-091) after adjuvant chemotherapy, **no PD-L1 restriction**: DFS HR 0.76 overall — **yet the PD-L1 ≥50% subgroup was not significant**, the reverse of what you would predict

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PERIOPERATIVE AND ADJUVANT DOSING

| Regimen                                         | Dosing                                                                                                                                                           |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CheckMate 816</b><br><i>neoadjuvant only</i> | Cisplatin <b>75 mg/m²</b> + pemetrexed <b>500 mg/m²</b> (or gemcitabine <b>1250 d1+8</b> ) + <b>nivolumab 360 mg</b> on day 1 — <b>q21d ×3</b> , then surgery    |
| <b>KEYNOTE-671</b><br><i>perioperative</i>      | Cisplatin 75 + pemetrexed 500 (or gemcitabine 1000 d1+8) + <b>pembrolizumab 200 mg</b> q21d ×4 → surgery → within 12 weeks, <b>pembrolizumab 200 mg q21d ×13</b> |
| <b>AEGEAN</b><br><i>perioperative</i>           | Cisplatin 75 + pemetrexed 500 (or gemcitabine 1250 d1+8) + <b>durvalumab 1500 mg</b> q21d ×4 → surgery → <b>durvalumab 1500 mg q28d ×12</b>                      |
| <b>CheckMate 77T</b><br><i>perioperative</i>    | Cisplatin 75 + pemetrexed 500 (or docetaxel 75) + <b>nivolumab 360 mg</b> q21d ×4 → surgery → within 90 days, <b>nivolumab 360 mg q28d ×1 year</b>               |
| <b>Adjuvant chemo</b>                           | Cisplatin <b>75 mg/m²</b> + pemetrexed 500 (non-squamous), gemcitabine 1250 d1+8, or docetaxel 75 — <b>q21d ×4</b> . Growth factor support if using docetaxel    |
| <b>Adjuvant IO</b>                              | <b>Atezolizumab 1680 mg q4wk</b> or <b>pembrolizumab 200 mg q3wk</b> , after adjuvant chemotherapy                                                               |

CHEMORADIATION AND METASTATIC DOSING

| Regimen                                   | Dosing                                                                                                                                                                                                                                  |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ChemoRT non-squamous</b>               | Carboplatin <b>AUC 5</b> + pemetrexed <b>500 mg/m²</b> q21d ×4, with daily RT                                                                                                                                                           |
| <b>ChemoRT squamous</b>                   | Cisplatin <b>50 mg/m² days 1, 8, 29, 36</b> + etoposide <b>50 mg/m² days 1–5 and 29–33</b> ; or carboplatin <b>AUC 2</b> + paclitaxel <b>45 mg/m²</b> weekly — with daily RT                                                            |
| <b>Consolidation durvalumab</b>           | <b>10 mg/kg q14d ×12 months</b> (or 1500 mg q4wk). <b>Start within 6 weeks</b> of finishing chemoRT                                                                                                                                     |
| <b>Platinum doublet + IO</b>              | Carboplatin <b>AUC 5–6</b> + pemetrexed 500 (non-squamous) or paclitaxel 200 (squamous) + <b>pembrolizumab 200 mg</b> q21d ×4 → pemetrexed + pembrolizumab maintenance                                                                  |
| <b>Single-agent IO</b>                    | Pembrolizumab <b>200 mg q3wk</b> (or 400 q6wk) · atezolizumab <b>1200 mg q3wk</b> · cemiplimab <b>350 mg q3wk</b>                                                                                                                       |
| <b>Osimertinib + chemo <i>FLAURA2</i></b> | Carboplatin <b>AUC 6</b> + pemetrexed <b>500</b> q21d ×4, with <b>osimertinib 80 mg daily continuously</b> , then pemetrexed maintenance                                                                                                |
| <b>Amivantamab + lazertinib</b>           | Amivantamab weight-based (<80 kg <b>1050 mg</b> , ≥80 kg <b>1400 mg</b> ) — split over days 1–2 in week 1, weekly through week 5, then <b>q2wk from week 7</b> ; + <b>lazertinib 240 mg PO daily</b> . <b>VTE prophylaxis ×4 months</b> |
| <b>Docetaxel, 2L</b>                      | <b>35 mg/m² days 1+8 q21d</b> — preferred over 75 q21d; reduce to 30 or 25 if heavily pretreated. <b>± ramucirumab 10 mg/kg</b>                                                                                                         |
| <b>Pemetrexed maintenance</b>             | <b>500 mg/m² q21d</b> — non-squamous only                                                                                                                                                                                               |

MONITORING AND CAUTIONS

- **Baseline HBsAg + anti-HBc before any immunosuppressive therapy**; consider quantitative TB testing by risk
- **Pemetrexed**: folic acid from ≥1 week before until 3 weeks after · B12 1000 mcg q9wk · dexamethasone 4 mg BID the day before and after, for rash. **Never in squamous**
- **Bevacizumab**: contraindicated in **squamous** histology, significant hemoptysis, and large cavitary tumors. Urine protein every other cycle; treat hypertension
- **Cisplatin**: avoid with hearing loss, CrCl <30, severe neuropathy, or poor performance status. Extensive hydration and aprepitant-containing antiemetics
- **Osimertinib**: echo at baseline and q3 months ×6 months; ECG at baseline, 2 weeks, 6 weeks, and with any QT-prolonging drug
- **Alectinib**: CK at baseline, 2 weeks, then q3 months · LFTs q2 weeks ×3 months. **Lorlatinib**: CNS effects and hyperlipidemia — median onset **15 days**; rosuvastatin first-line
- **Immunotherapy**: TSH and free T4 at baseline then q4–6 weeks. Prednisone **1–1.5 mg/kg** for irAEs with a slow taper
- **Do not start a targeted agent — osimertinib and crizotinib above all — within 3 months of immunotherapy**; the toxicity is additive. IO is also less effective in EGFR/ALK-driven disease

KEY TRIALS

|                         |                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>CheckMate 816</b>    | Neoadjuvant platinum + nivolumab ×3 — pCR 24% vs 2%, EFS HR 0.63, OS HR 0.62                                |
| <b>KEYNOTE-671</b>      | Perioperative pembrolizumab — EFS HR 0.58; the first perioperative IO trial with a confirmed OS benefit     |
| <b>AEGEAN / 77T</b>     | Perioperative durvalumab (EFS HR 0.68) / nivolumab (EFS HR 0.58)                                            |
| <b>ADAURA / ALINA</b>   | Adjuvant osimertinib ×36 mo (DFS HR 0.17, OS HR 0.49) / alectinib ×24 mo (DFS HR 0.24)                      |
| <b>PACIFIC</b>          | Consolidation durvalumab after chemoRT — 5-yr OS <b>43% vs 33%</b>                                          |
| <b>LAURA</b>            | Consolidation osimertinib, EGFR-mutant stage III — PFS <b>39.1 vs 5.6 mo</b> , HR 0.16                      |
| <b>FLAURA / FLAURA2</b> | Osimertinib 1L; adding chemo gives mPFS 25.5 vs 16.7 mo and OS 47.5 vs 37.6                                 |
| <b>MARIPOSA</b>         | Amivantamab + lazertinib vs osimertinib — mPFS 23.7 vs 16.6 mo, <b>OS HR 0.75</b> , at a real toxicity cost |
| <b>CROWN</b>            | Lorlatinib 1L in ALK — <b>5-yr PFS 60% vs 8%</b>                                                            |
| <b>PAPILLON</b>         | Amivantamab + chemo 1L for EGFR exon 20 — mPFS 11.4 vs 6.7 mo                                               |
| <b>LIBRETTO-431</b>     | Selpercatinib vs chemo ± pembrolizumab in RET — mPFS 24.8 vs 11.2 mo                                        |
| <b>CALGB 140503</b>     | Sublobar resection non-inferior for peripheral cT1aN0 ≤2 cm — DFS HR 1.01                                   |